Operation Management

BITS Pilani
Pilani Campus

Abhishek Maiti

BITS Pilani
Pilani Campus

< Operation Management>
Lecture No. 9

Inventory Management

- A stock or store of goods.
- It is to strike a balance between inventory investment and customer service.

Functions of Inventory

- To meet anticipated customer demand.
- To smooth production requirements.
- To decouple operations.
- To reduce the risk of stock-outs.
- To take advantage of order cycles.
- To hedge against price increases.
- To permit operations.

Types of Inventory

Raw material inventory : Materials that are usually purchased but have yet to enter the manufacturing process.

Work-in-process (WIP) inventory : Products or components that are no longer raw materials but have yet to become finished products.

Maintenance/repair/operating (MRO) inventory: Maintenance, repair, and operating materials.

Finished-goods inventory : An end item ready to be sold, but still an asset on the company's books.

Managing the Inventory

- How accurate inventory records can be maintained (Inventory Counting System).
- How inventory items can be classified (called *ABC analysis*).

Inventory Counting System

- **Periodic system :** Physical count of items in inventory made at periodic intervals (weekly, monthly).
- **Two-bin system :** Two containers of inventory; reorder when the first is empty.
- **Perpetual inventory system:** System that keeps track of removals from inventory continuously, thus monitoring current levels of each item.

Inventory Counting System

- **Universal product code (UPC) :** Bar code printed on a label that has information about the item to which it is attached.
- **Point-of-sale (POS) systems :** Record items at time of sale.

ABC analysis

- A method for dividing on-hand inventory into three classifications based on annual dollar volume.
- Class A items (very important),
- Class B items (moderately important)
- Class C items (least important).

Class A items

- Class A items are the large investment items; vital few.
- 15-20 percent of the items accounting for 75-80 percent of the total annual usage value.
- Computerized Materials Requirement Planning (MRP) would be beneficial.

Class B items

- 25-30 percent of the items accounting for 15-25 percent of the total annual usage value.
- Historical data based order rather applying forecasting techniques.
- Economic Order Quantity (EOQ) would be beneficial.

Class C items

- Class C items are the large investment items; trivial many.
- 50 percent of the items accounting for 5-10 percent of the total annual usage value.
- Quantity discount model would be beneficial.

Example 1

LO 12.1 *Conduct* an ABC analysis

ABC ANALYSIS FOR A CHIP MANUFACTURER

Silicon Chips, Inc., maker of superfast DRAM chips, wants to categorize its 10 major inventory items using ABC analysis.

APPROACH ► ABC analysis organizes the items on an annual dollar-volume basis. Shown below (in columns 1–4) are the 10 items (identified by stock numbers), their annual demands, and unit costs.

SOLUTION ► Annual dollar volume is computed in column 5, along with the percentage of the total represented by each item in column 6. Column 7 groups the 10 items into A, B, and C categories.

ABC Calculation

(1) ITEM STOCK NUMBER	(2) PERCENTAGE OF NUMBER OF ITEMS STOCKED	(3) ANNUAL VOLUME (UNITS)	×	(4) UNIT COST	=	(5) ANNUAL DOLLAR VOLUME	(6) PERCENTAGE OF ANNUAL DOLLAR VOLUME		(7) CLASS
#10286	20%	1,000		$ 90.00		$ 90,000	38.8%	72%	A
#11526		500		154.00		77,000	33.2%		A
#12760	30%	1,550		17.00		26,350	11.3%	23%	B
#10867		350		42.86		15,001	6.4%		B
#10500		1,000		12.50		12,500	5.4%		B
#12572	50%	600		14.17		8,502	3.7%	5%	C
#14075		2,000		.60		1,200	.5%		C
#01036		100		8.50		850	.4%		C
#01307		1,200		.42		504	.2%		C
#10572		250		.60		150	.1%		C
		8,550				$232,057	100.0%		

INSIGHT ► The breakdown into A, B, and C categories is not hard and fast. The objective is to try to separate the "important" from the "unimportant."

LEARNING EXERCISE ► The unit cost for Item #10286 has increased from $90.00 to $120.00. How does this impact the ABC analysis? [Answer: The total annual dollar volume increases by $30,000, to $262,057, and the two A items now comprise 75% of that amount.]

RELATED PROBLEMS ► 12.1, 12.2, 12.3 (12.5–12.6 are available in MyOMLab)

EXCEL **OM** Data File **Ch12Ex1.xls** can be found in MyOMLab.

Guesstimation

How much Accuracy is Needed

- APICS recommends the following guidelines for inventory record accuracy:
- 0.2 percent for A items, 1 percent for B items and 5 percent for C items.

Inventory Costs

- Purchase cost
- Holding (carrying) cost
- Ordering costs
- Setup costs
- Shortage costs

Any Question

Inventory Models

- Independent Demand Model.
- Dependent Demand Model. (Will be deal later in Dependent Demand Management)

Independent Demand Model

1. Basic economic order quantity (EOQ) model.

2. Production order quantity model.

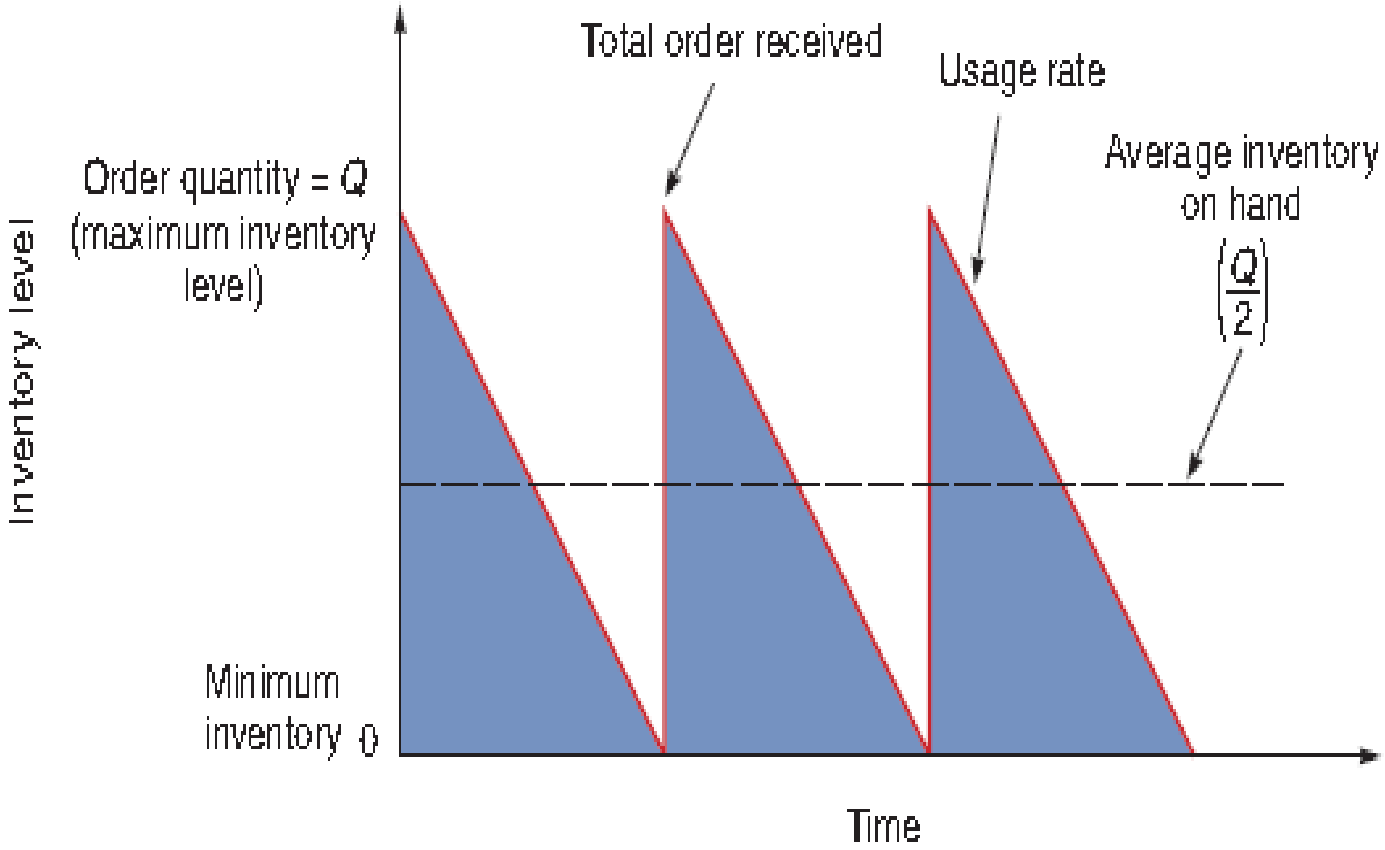
3. Quantity discount model.

EOQ Model

- EOQ model is the simplest of the three models.
- An inventory-control technique that minimizes the total of ordering and holding costs.

Source: Operation Management by William J Stevenson

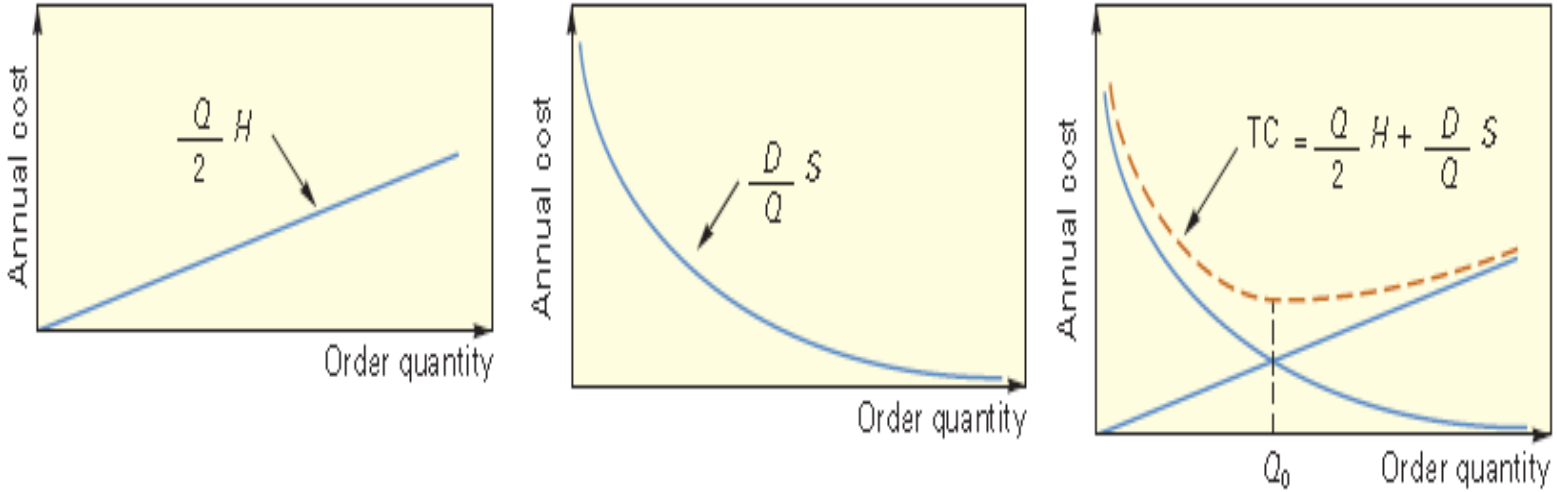
EOQ Model



Source: Heizer, Render, Muckson

FIGURE 13.4

Carrying cost, ordering cost, and total cost curve



A. Carrying costs are linearly related to order size.

B. Ordering costs are inversely and nonlinearly related to order size.

C. The total-cost curve is U-shaped.

Source: Operation Management by William J Stevenson

Minimizing Costs

1. Annual setup cost = (Number of orders placed per year) × (Setup or order cost per order)

$$= \left(\frac{\text{Annual demand}}{\text{Number of units in each order}}\right)(\text{Setup or order cost per order})$$

$$= \left(\frac{D}{Q}\right)(S) = \frac{D}{Q}S$$

2. Annual holding cost = (Average inventory level) × (Holding cost per unit per year)

$$= \left(\frac{\text{Order quantity}}{2}\right)(\text{Holding cost per unit per year})$$

$$= \left(\frac{Q}{2}\right)(H) = \frac{Q}{2}H$$

3. Optimal order quantity is found when annual setup (order) cost equals annual holding cost, namely:

$$\frac{D}{Q}S = \frac{Q}{2}H$$

4. To solve for Q^*, simply cross-multiply terms and isolate Q on the left of the equal sign:

$$2DS = Q^2H$$

$$Q^2 = \frac{2DS}{H}$$

$$Q^* = \sqrt{\frac{2DS}{H}} \qquad (12\text{-}1)$$

Minimizing Costs

Q = Number of units per order

Q^* = Optimum number of units per order (EOQ)

D = Annual demand in units for the inventory item

S = Setup or ordering cost for each order

H = Holding or carrying cost per unit per year

$$\text{Expected number of orders} = N = \frac{\text{Demand}}{\text{Order quantity}} = \frac{D}{Q^*} \qquad (12\text{-}2)$$

$$\text{Expected time between orders} = T = \frac{\text{Number of working days per year}}{N} \qquad (12\text{-}3)$$

Minimizing Costs

$$\text{Total annual cost} = \text{Setup (order) cost} + \text{Holding cost} \tag{12-4}$$

In terms of the variables in the model, we can express the total cost TC as:

$$TC = \frac{D}{Q}S + \frac{Q}{2}H \tag{12-5}$$

$$TC = \frac{D}{Q}S + \frac{Q}{2}H + PD$$

Class Exercise

- Sharp, Inc., a company that markets painless hypodermic needles to hospitals, would like to reduce its inventory cost by determining the optimal number of hypodermic needles to obtain per order.
- The annual demand is 1,000 units; the setup or ordering cost is $10 per order; and the holding cost per unit per year is $.50.
- What is the EOQ ?

Class Exercise

- Sharp, Inc. has a 250-day working year and wants to find the number of orders (N) and the expected time between orders (T).

Class Exercise

- Sharp, Inc. wants to determine the combined annual ordering and holding costs.

Re-Order Point

Lead time :

- In purchasing systems, the time between placing an order and receiving it.
- In production systems, the wait, move, queue, setup, and run times for each component produced.

Reorder point (ROP) : The inventory level (point) at which action is taken to replenish the stocked item.

Re-Order Point

- **Safety stock (*ss*)** : Extra stock to allow for uneven demand; a buffer.
- ROP = Demand per day * Lead time for a new order in days
- ROP = $d * L$
- ROP = Expected demand during lead time + Safety stock.
- $d = \dfrac{D}{No.of\ working\ days\ in\ a\ year}$
- d = demand per day, D= Annual demand.

Class Exercise

- An Apple store has a demand (D) for 8,000 iPhones per year.
- The firm operates a 250-day working year.
- On average, delivery of an order takes 3 working days, but has been known to take as long as 4 days.
- The store wants to calculate the reorder point without a safety stock and then with a one-day safety stock.

Any Question

Production Order Quantity Model

- Also known as Economic Production Quantity.
- The assumptions of the EPQ model are similar to those of the EOQ model, except that instead of orders received in a single delivery, units are received incrementally during production.

$$
\begin{aligned}
Q &= \text{Number of units per order} \\
H &= \text{Holding cost per unit per year} \\
p &= \text{Daily production rate} \\
d &= \text{Daily demand rate, or usage rate} \\
t &= \text{Length of the production run in days}
\end{aligned}
$$

EPQ Model

1. $\begin{pmatrix}\text{Annual inventory}\\ \text{holding cost}\end{pmatrix} = (\text{Average inventory level}) \times \begin{pmatrix}\text{Holding cost}\\ \text{per unit per year}\end{pmatrix}$

2. $(\text{Average inventory level}) = (\text{Maximum inventory level})/2$

3. $\begin{pmatrix}\text{Maximum}\\ \text{inventory level}\end{pmatrix} = \begin{pmatrix}\text{Total production during}\\ \text{the production run}\end{pmatrix} - \begin{pmatrix}\text{Total used during}\\ \text{the production run}\end{pmatrix}$

$$= pt - dt$$

However, Q = total produced $= pt$, and thus $t = Q/p$. Therefore:

$$\text{Maximum inventory level} = p\left(\frac{Q}{p}\right) - d\left(\frac{Q}{p}\right) = Q - \frac{d}{p}Q$$
$$= Q\left(1 - \frac{d}{p}\right)$$

4. Annual inventory holding cost (or simply holding cost) =

$$\frac{\text{Maximum inventory level}}{2}(H) = \frac{Q}{2}\left[1 - \left(\frac{d}{p}\right)\right]H$$

EPQ Model

$$\text{Setup cost} = (D/Q)S \qquad \text{Holding cost} = \tfrac{1}{2}HQ[1 - (d/p)]$$

Set ordering cost equal to holding cost to obtain Q_p^*:

$$\frac{D}{Q}S = \tfrac{1}{2}HQ[1 - (d/p)]$$

$$Q^2 = \frac{2DS}{H[1 - (d/p)]}$$

$$Q_p^* = \sqrt{\frac{2DS}{H[1 - (d/p)]}} \qquad (12\text{-}7)$$

- $$d = \frac{D}{No.of\ days\ the\ plant\ is\ in\ operation}$$

$$Q_p^* = \sqrt{\frac{2DS}{H\left(1 - \frac{\text{Annual demand rate}}{\text{Annual production rate}}\right)}} \qquad (12\text{-}8)$$

Class Exercise

- Nathan Manufacturing, Inc., makes and sells specialty hubcaps for the retail automobile aftermarket.
- Nathan's forecast for its wire-wheel hubcap is 1,000 units next year, with an average daily demand of 4 units.
- However, the production process is most efficient at 8 units per day.
- So the company produces 8 per day but uses only 4 per day.
- The company wants to solve for the optimum number of units per order.

Quantity Discount Model

- A reduced price for items purchased in large quantities.

Total annual cost = Annual setup (ordering) cost + Annual holding cost + Annual product cost,

or

$$TC = \frac{D}{Q}S + \frac{Q}{2}IP + PD \qquad (12\text{-}9)$$

where Q = Quantity ordered
D = Annual demand in units
S = Setup or ordering cost per order
P = Price per unit
I = Holding cost per unit per year expressed as a percent of price P

$$Q^* = \sqrt{\frac{2DS}{IP}}$$

Class Exercise

- Chris Beehner Electronics stocks toy remote control flying drones.
- Recently, the store has been offered a quantity discount schedule for these drones.
- Furthermore, setup cost is $200 per order, annual demand is 5,200 units, and annual inventory carrying charge as a percent of cost, I , is 28%.
- What order quantity will minimize the total inventory cost?

TABLE 12.2 A Quantity Discount Schedule

PRICE RANGE	QUANTITY ORDERED	PRICE PER UNIT *P*
Initial price	1–119	$100
Discount price 1	120–1,499	$ 98
Discount price 2	1,500 and over	$ 96

Solution Procedure

STEP 1: Starting with the *lowest* possible purchase price in a quantity discount schedule and working toward the highest price, keep calculating Q^* until the first feasible EOQ is found.

STEP 2: Calculate the total annual cost TC for each of the possible best order quantities determined in Step 1.

Select the quantity that has the lowest total cost.

Solution

TABLE 12.3 Total Cost Computations for Chris Beehner Electronics

ORDER QUANTITY	UNIT PRICE	ANNUAL ORDERING COST	ANNUAL HOLDING COST	ANNUAL PRODUCT COST	TOTAL ANNUAL COST
275	$98	$3,782	$ 3,773	$509,600	$517,155
1,500	$96	$ 693	$20,160	$499,200	$520,053

Source: Heizer, Render, Muckson

Any Question

Probabilistic Models and Safety Stock

- Probabilistic model : A statistical model applicable when product demand or any other variable is not known but can be specified by means of a probability distribution.
- Reorder point = ROP = $d * L$
- d = Daily demand
- L = Order lead time, or number of working days it takes to deliver an order
- ROP = $d * L + ss$

Take Home Exercise

- Example 12 of your Text Book.

Other Probabilistic Model

1. Demand is variable, and lead time is constant.

2. Lead time is variable, and demand is constant.

3. Both demand and lead time are variable.

Demand is Variable and Lead time is constant

$$\text{ROP} = (\textit{Average}\ \text{daily demand} \times \text{Lead time in days}) + Z\sigma_{dLT} \qquad (12\text{-}15)$$

where σ_{dLT} = Standard deviation of demand during lead time = $\sigma_d\sqrt{\text{Lead time}}$

and σ_d = Standard deviation of demand per day

- The *average* daily demand for Lenovo laptop computers at a Circuit Town store is 15, with a standard deviation of 5 units
- The lead time is constant at 2 days. Find the reorder point if management wants a 90% service level (i.e., risk stockouts only 10% of the time).
- How much of this is safety stock?

Lead time is Variable and Demand is constant

Lead Time Is Variable and Demand Is Constant When the demand is constant and *only the lead time is variable*, then:

$$\text{ROP} = (\text{Daily demand} \times \textit{Average}\ \text{lead time in days}) + Z \times \text{Daily demand} \times \sigma_{LT} \quad (12\text{-}16)$$

where σ_{LT} = Standard deviation of lead time in days

- The Circuit Town store sells about 10 digital cameras a day (almost a constant quantity).
- Lead time for camera delivery is normally distributed with a mean time of 6 days and a standard deviation of 1 day. A 98% service level is set.
- Find the ROP.

Both Demand & Lead time is Variable

Both Demand and Lead Time Are Variable When both the demand and lead time are variable, the formula for reorder point becomes more complex:[6]

$$\text{ROP} = (\text{Average daily demand} \times \text{Average lead time in days}) + Z\sigma_{dLT} \qquad (12\text{-}17)$$

where

$$\sigma_d = \text{Standard deviation of demand per day}$$
$$\sigma_{LT} = \text{Standard deviation of lead time in days}$$

$$\text{and } \sigma_{dLT} = \sqrt{(\text{Average lead time} \times \sigma_d^2) + (\text{Average daily demand})^2\sigma_{LT}^2}$$

- The Circuit Town store's most popular item is six-packs of 9-volt batteries.
- About 150 packs are sold per day, following a normal distribution with a standard deviation of 16 packs.
- Batteries are ordered from an out-of-state distributor; lead time is normally distributed with an average of 5 days and a standard deviation of 1 day.
- To maintain a 95% service level, what ROP is appropriate?

Single Period Inventory

- Single-period inventory model: A system for ordering items that have little or no value at the end of a sales period (perishables).
- Service level : The probability that demand will not be greater than supply during lead time. It is the complement of the probability of a stock out.

C_s = Cost of shortage (we underestimated) = Sales price per unit − Cost per unit

C_o = Cost of overage (we overestimated) = Cost per unit − Salvage value per unit (if there is any)

The service level, that is, the probability of *not* stocking out, is set at:

$$\text{Service level} = \frac{C_s}{C_s + C_o} \qquad (12\text{-}18)$$

Source: Heizer, Render, Muckson

Take Home Exercise

- Chris Ellis's newsstand, just outside the Smithsonian subway station in Washington, DC, usually sells 120 copies of the *Washington Post* each day.
- Chris believes the sale of the *Post* is normally distributed, with a standard deviation of 15 papers. He pays 70 cents for each paper, which sells for $1.25.
- The *Post* gives him a 30-cent credit for each unsold paper.
- He wants to determine how many papers he should order each day and the stock out risk for that quantity.
- Example 15

Fixed Period System

- A system in which inventory orders are made at regular time intervals.
- A fixed-period (*P*) system is appropriate when vendors make routine (i.e., at fixed-time interval) visits to customers to take fresh orders or when purchasers want to combine orders to save ordering and transportation costs.

- **Inventory counted only at end of period**
- **Order brings inventory up to target level**
  - Only relevant costs are ordering and holding
  - Lead times are known and constant
  - Items are independent of one another

Fixed-Period Systems

- Inventory is only counted at each review period
- May be scheduled at convenient times
- Appropriate in routine situations
- May result in stockouts between periods
- May require increased safety stock

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 12
2. Operation Management by William J Stevenson—Chapter 13
3. Application of ABC analysis in Inventory Management, Technical Note, IIM Ahmedabad.

THANK YOU